

IERG3840 Web Application Development Project (2025-26 Term 1)

Lab 1 Specification

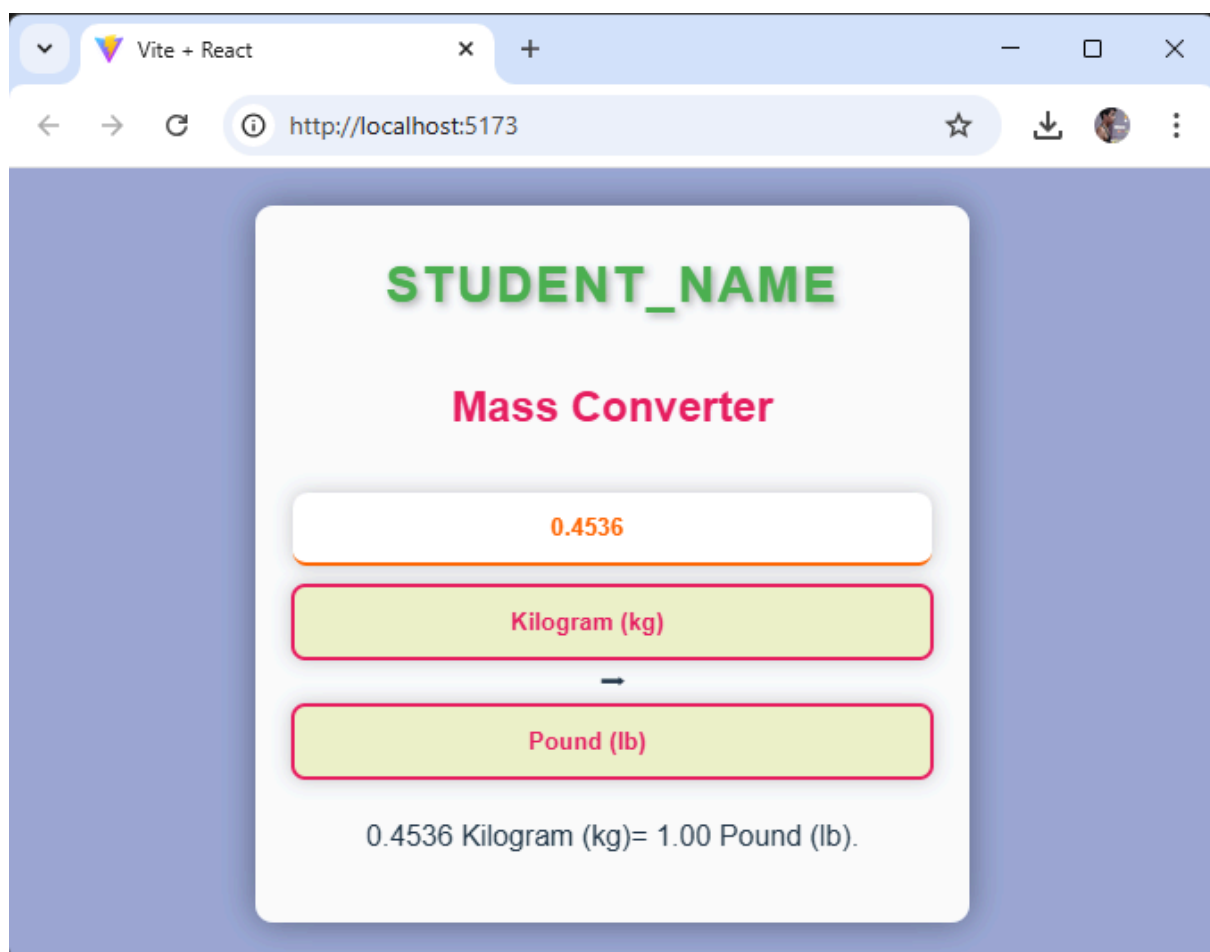
Lab 1 is to design a mass converter interface for your web service using React JS in your laptop. After understanding the temperature converter demonstration in the lecture note, update it to the mass converter web service. It will take 15% of the course total.

Part 1: Demonstration (7%)

Date: 18th Sept, 2025 (Thur)

Time and Venue: In Class

You need to bring your own laptop for demonstration



In your demonstration, you need to demonstrate the following features:

- Show your full name (1%)
- Users can select three different units in the interface, which are Kilogram (kg), Pound (lb) and the assigned unit based on the last digit of your student ID. (2%)

Last digit of your student ID	Student ID example	Your assigned unit	Short form of the unit	1 lb =
0 and 1	1155000000 1155000001	Gram	g	453.6 g
2 and 3	1155000002 1155000003	Ounce	oz	16 oz
4 and 5	1155000004 1155000005	Stone	st	1/14 st
6 and 7	1155000006 1155000007	US Ton	Short Ton	1/2000 Short Ton
8 and 9	1155000008 1155000009	Imperial Ton	Long Ton	1/2240 Long Ton

- When users type the number in the input box, the currency converter functions properly. (2%)
 - The exchange rate for kilogram and pound: **0.4536 kg = 1 lb**
 - The exchange rate for pound and your assigned unit: see the table above
 - The displayed result is corrected to two decimal places
- When users select the same unit, you need to show an error message: **Conversion between the same unit is not allowed.** (1%)
- When users type non-numeric values into the input box, you need to show an error message: **Enter a valid numeric value.** (1%)

Part 2: Question and Answer (5%)

You will be asked one set of problems out of the five sets below randomly after completing the demonstration. You need to answer the problem set verbally.

You can prepare a cheat sheet for the Q&A session, however, you cannot search online or search from the lecture notes during the Q&A session.

There are five points in each set.

Set 1:

- (a) What is the name of the icon on the tab? (1%)
- (b) What is a markup language? (2%)
- (c) What is the most recent html version? (1%)
- (d) What is an inline element in html? (1%)

Set 2: This problem set is about JavaScript.

- (a) What is the most recent JavaScript version now? (1%)
- (b) What does null mean in JavaScript? (1%)
- (c) What does undefined mean in JavaScript? (1%)
- (d) What is a template literal in JavaScript? (2%)

Set 3: This problem set is about JavaScript.

- (a) What are var, let and const mean in JavaScript? (3%)
- (b) Suggest one non-primitive data type in JavaScript, and explain what non-primitive data type is using your example. (2%)

Set 4: This problem set is about JavaScript.

- (a) What is string coercion in JavaScript? (1%)
- (b) What is boolean coercion in JavaScript? (1%)
- (c) Explain why JavaScript is a dynamically typed language. (1%)
- (d) What does this mean when this is used in the global context? (1%)
- (e) What does this mean when this is used in the event handlers? (1%)

Set 5: This problem set is about JavaScript.

- (a) In Javascript, what are arrow functions? (2%)
- (b) Give a major difference between normal functions and arrow functions in Javascript. (1%)
- (c) Suggest and explain two major reasons why arrow functions are used extensively in web/mobile applications? (2%)

Part 3: Short Report (3%)

Use the template, fill in all the necessary information and submit the **PDF** format on or before 19th Sept, 2025 (Fri). Without demonstration, no scores will be given for the submitted report.

Late Policy

The following behaviours are regarded as late:

- Demonstration and Q&A are not performed on the specified date without justification
- Short report is submitted after the deadline

Late penalties will be counted on the late parts only.

- Demonstration and Q&A: 30% off for late within one week
- Report: 30% off for late within one week
- No scores after one week

< End of Specification >